SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology
Department of Agricultural Engineering

AGE 219: Basics of Computer Programming

Assignment 2: Advanced Data Handling and Functional Automation

Covering Modules 4, 5, and 6

Course Instructor:

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Submission Deadline: Exactly one week after the completion of Lecture 6
Submission Format: Exported PDF of Google Colab Notebook
Academic Integrity: See Honest Clause in Syllabus

This assignment evaluates your mastery of iterative logic, modular programming, and data organization. You are required to design reusable engineering tools and process batch sensor data efficiently.

The Case Scenario: The Precision Agri-Intelligence Hub

Following your success in the first phase, the **Sokoine Agri-Tech Research Block (SARB)** has expanded. They have deployed a network of multi-parameter sensors and a modern fleet of tractors equipped with telematics.

The management now requires a more sophisticated software layer. You must build functions that can calculate complex hydraulic parameters, loops that can process entire days of environmental logs, and data structures that can manage a multi-machine inventory.

Part A: Iteration and Loops

Question 1: Cumulative Rainfall Monitor (10 Marks)

A list contains 10 days of rainfall data in mm: `daily_rain = [0.0, 15.2, 8.5, 0.0, 0.0, 22.1, 4.0, 0.0, 12.5, 5.0]`. **Task:** Write a script using a `for` loop to calculate the cumulative rainfall at the end of each day. Print the result for each day as: "Day X: Cumulative Rain = Y mm".

Question 2: Sensor Peak and Average Analysis (10 Marks)

A data logger recorded 100 greenhouse temperatures (simulated as a list of random floats between 18.0 and 35.0). **Task:** Use a loop (do not use the `max()` function) to find the highest temperature in the list. Also, calculate the average temperature of the entire set using a loop-based accumulator.

Question 3: Grid Calibration (10 Marks)

A 3x3 grid of soil moisture sensors is represented as a list of lists:

```
moisture_grid = [
    [25, 28, 22],
    [30, 24, 26],
    [21, 23, 22]
]
```

Task: Use **Nested Loops** to print the coordinates and value of every sensor that reports moisture below 25%. Format: "Low Moisture at (Row, Col): Value%".

Part B: Functions and Modular Programming

Question 4: Hydraulic Design Module (15 Marks)

Write a function `calculate_trapezoidal_flow(b, h, z, v)` that calculates the flow rate (Q) in a trapezoidal irrigation canal.

- **Formula:** $Q = Area \times Velocity$.
- **Area Calculation:** $Area = (b + z \cdot h) \cdot h$.
- **Parameters:** b (bottom width), h (depth), z (side slope), v (velocity).

Ensure the function has a proper **Docstring** and returns the flow rate in m^3/s. Test it with $b = 2.5$, $h = 1.2$, $z = 1.5$, and $v = 0.8$.

Question 5: Recursive Unit Conversion (10 Marks)

Engineers often convert Application Rates. Write a function `convert_rate(val, to_metric=True)`:

- If `to_metric` is `True`, convert from Pounds per Acre (lb/ac) to Kilograms per Hectare (kg/ha). (Factor: 1.12085).
- If `False`, convert the other way.

Include a default argument for `to_metric`.

Part C: Data Structures (Lists, Tuples, Dictionaries)

Question 6: Multi-Parameter Mapping (10 Marks)

Create a dictionary named `iot_node_metadata` where the Keys are unique node IDs (e.g., "NODE_01"). The Values should be **Tuples** containing (Latitude, Longitude, SensorType). **Task:** Write a script that iterates through the dictionary and prints only the IDs of nodes that are "Soil Moisture" sensors.

Question 7: List Comprehension for Data Cleaning (10 Marks)

A list of sensor readings contains "Noise" in the form of negative values which are physically impossible. `raw_readings = [22.5, -99.0, 23.1, 24.8, -1.0, 22.9, 0.0]` **Task:** Use a **List Comprehension** to create a new list `clean_readings` that excludes all negative values. Then, use another list comprehension to convert all values in `clean_readings` to a percentage (assuming a max value of 50).

Question 8: Machinery Fleet Management (15 Marks)

Represent a fleet of 3 tractors using a **List of Dictionaries**. Each dictionary must have keys: `id`, `hp`, `hours`, and `is_working`. **Task:** Write a function that takes this fleet list as an argument and returns the `id` of the tractor with the highest number of operating hours.

Part D: Integrated Engineering Systems

Question 9: The Automated Grain Batch Processor (15 Marks)

Write a program that simulates a conveyor belt processing 10 batches of grain.

1. Create a function `grade_batch(moisture)` that returns "PASS" if moisture < 14 and "FAIL" otherwise.
2. Create a list of 10 random moisture values.
3. Use a `for` loop to process the list. Use a `while` loop inside to "re-test" any batch that fails (decrement moisture by 0.5 until it passes).
4. Store the final results in a dictionary where the keys are "Batch 1", "Batch 2", etc.

Question 10: Debugging Case Study (10 Marks)

The following function is supposed to calculate the total fuel needed for a list of tractor tasks, but it has multiple errors. Identify the errors and provide the corrected version.

```
def get_total_fuel(tasks_list)
    total = 0
    for task in task_list:
        total = total + task["fuel"]
    return Fuel is total

my_tasks = [{"id":1, "fuel":20}, {"id":2, "fuel":15}]
print(get_total_fuel(my_tasks))
```

Question 11: The System Report (10 Marks)

Use **f-strings** to generate a professional multi-line report. The report must pull data from a dictionary representing a farm block, including the list of crops grown and the calculated total area (summing up sub-plots).

End of Assignment 2

Department of Agricultural Engineering, SUA